Adjective Ordering Preferences: An Incremental RSA Account with Bayesian Inference

LMBayes Meeting 2026 — Discussion Handout

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March 5, 2026 Toward (finally and hopefully!!) a first draft of the full paper

Introduction

Speakers of diverse languages exhibit robust ordering preferences of adjectives (*e.g.*, preferring "the big brown bag" over "the brown big bag"). One way to approach this regularity is through communicative efficiency: helping interlocutors coordinate quickly and reliably. Yet "efficiency" can be formalized from two perspectives with opposing predictions: a *hierarchical, comprehension-oriented* view in which modification composes outward from the noun (placing efficient adjectives closer to it), and a *linear, production-oriented* view in which speakers benefit from placing efficient adjectives early (putting them further from the noun). The present project pursues a unified perspective: adjective ordering is explained by *incremental pragmatic reasoning*, where a speaker produces one adjective at a time from left to right and the contextual informativeness of each adjective interacts with its inherent semantic imprecision through right-to-left hierarchical structure and compositional semantics.

The project has three main building blocks:

1. **Model.** An incremental variant of the Rational Speech Act (RSA) model, implemented in NumPyro / JAX, with Bayesian model comparisons.
2. **Experiments.** Two complementary experiments in referential communication—a slider-rating experiment, and a free-production experiment.
3. **Simulation.** A broad parameter sweep showing that the size-first advantage emerges without being assumed a priori, generalising our findings beyond the limited set of experiment items.

Discussion: High-level research questions

- What is the minimal pragmatic mechanism that predicts adjective ordering as an emergent consequence of communication, rather than stipulating a fixed order?
- What role does the interaction between *inherent* semantic imprecision and *contextual* informativeness play in shaping ordering preferences?
- Does incremental (left-to-right) production planning add explanatory power beyond what compositional semantics alone provides?

1 Model

1.1 Representational primitives

State space. An **object** is a triple $o = (s, c, f)$ where $s \in [1, 10]$ is size (used to construct experiment items), $c \in \{0, 1\}$ indicates colour match, and $f \in \{0, 1\}$ indicates form match. A **context** $\mathcal{C}$ is a set of n_{obj} objects; the referent is always the largest object with $c = f = 1$.

Utterance space. Utterances are built from two semantic types: S (size) and C (colour/form; form is subsumed under colour because they share the same determinate semantics). An utterance $u = (a_1, \ldots, a_T)$ is an ordered sequence of adjectives $a_t \in \{\text{S}, \text{C}, \text{F}\}$, where $T \in \{1, 2, 3\}$ is the utterance length. Combining lengths and orderings yields 15 surface forms: {S, SC, SCF, SF, SFC, C, CS, CSF, CF, CFS, F, FS, FSC, FC, FCS}. In all analyses, F is treated identically to C.

1.2 Meaning functions

Context-dependent, threshold-based size (S) semantics. We adopt a *slack semantics* based on the Approximate Number System. A context-dependent threshold is computed as

$$\theta_k(\mathcal{C}) = x^{\text{mid}}_{\text{max}}(\mathcal{C}) - k\left(x^{\text{mid}}_{\text{max}}(\mathcal{C}) - x^{\text{mid}}_{\text{min}}(\mathcal{C})\right), \tag{1}$$

where $x^{\text{mid}}_{\text{min}}$ and $x^{\text{mid}}_{\text{max}}$ are the q_{low}- and q_{high}-quantiles of the contextual size distribution ($q_{\text{low}} = 0.2$, $q_{\text{high}} = 0.8$). The semantic value of applying S to object o is then

$$[\![\text{S}]\!](o, \mathcal{C}) = \Phi\left(\frac{s_o - \theta_k(\mathcal{C})}{w_f\sqrt{s_o^2 + \theta_k(\mathcal{C})^2}}\right), \tag{2}$$

where θ_k is the threshold from eq. (1), Φ is the standard normal CDF, and $w_f \geq 0$ is a *perceptual blur* parameter.

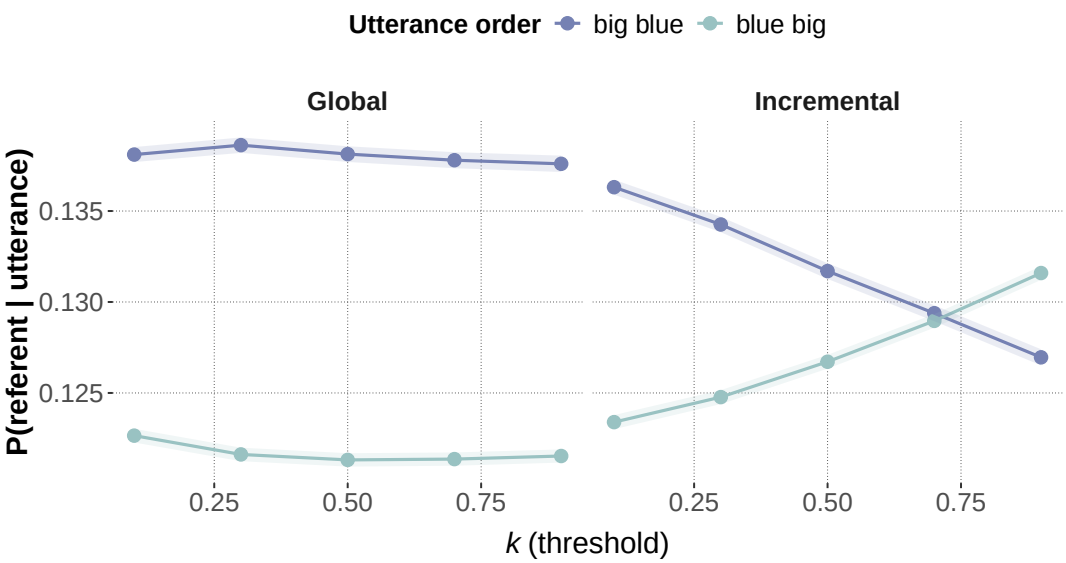

Figure 1: Effect of threshold k on communicative success (simulation, marginalised over all other parameters). Lower k (stricter threshold) reduces both orderings' success but narrows the gap between them. The incremental speaker maintains a consistent size-first advantage across k values.

Colour/Form (C, F). Both colour and form have determinate denotations: an object either matches the described colour/form or it does not ($c, f \in \{0, 1\}$). Their semantic value is parameterised by a scalar $\nu \in (0, 1)$, interpretable as the probability that a feature-matching object is correctly

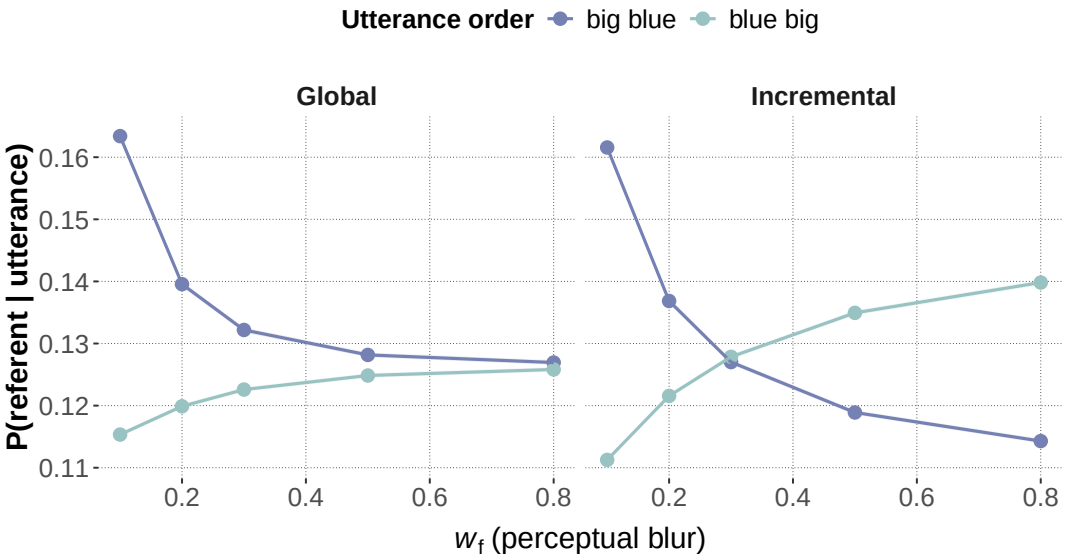

Figure 2: Effect of perceptual blur w_f on communicative success (simulation, marginalised over all other parameters). Higher w_f degrades the informativeness of size, reducing the advantage of big blue over blue big. The incremental speaker shows a larger separation between orderings than the global speaker.

described by the adjective—capturing residual uncertainty in the mapping from perceptual feature to word. In the slider model both C and F share a single ν; in the production model they receive separate values (ν_{colour}, ν_{form}), but both are fixed rather than inferred (see section 1.6).

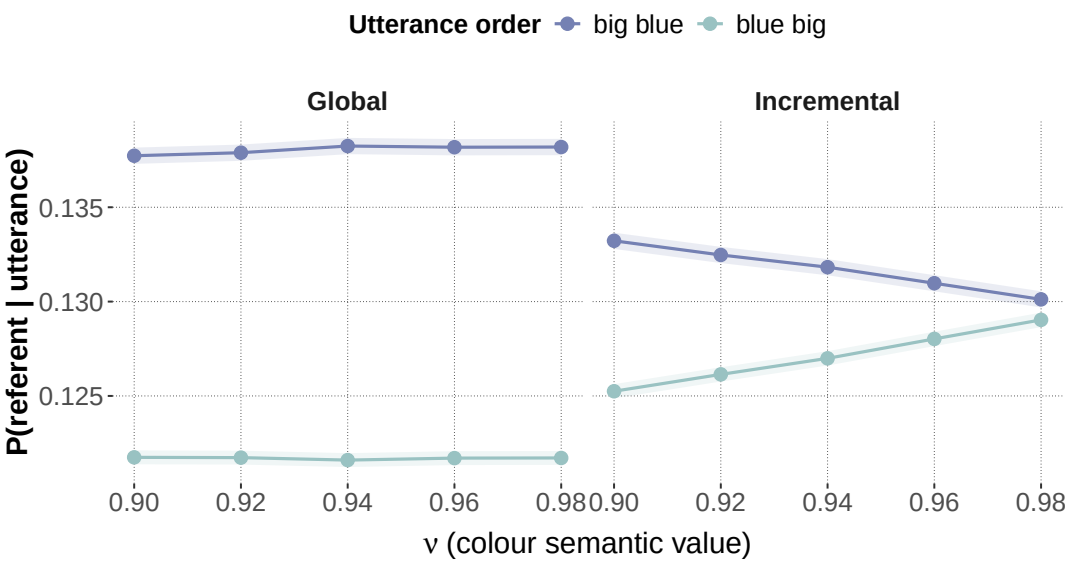

Figure 3: Effect of colour semantic value ν on communicative success (simulation, marginalised over all other parameters). Higher ν (more deterministic colour semantics) increases blue big success, narrowing the gap. The incremental speaker still favours size-first ordering across all ν values tested.

1.3 Incremental compositional semantics

Meanings compose *right to left*: processing begins at the adjective closest to the noun (a_T) and proceeds outward ($a_{T-1}, \ldots, a_1$), so that each successive adjective narrows the listener's posterior over the set of objects compatible with the already-interpreted suffix. Starting from a uniform prior

$p_T(o)$, each step applies the next adjective (scanning from $t = T$ down to $t = 1$):

$$p_{t-1}(o) \propto p_t(o) \cdot [\![a_t]\!](o, \mathcal{C}, p_t). \tag{3}$$

The *literal listener* $L_0(o \mid u, \mathcal{C})$ is the resulting posterior p_0 after all adjectives in u have been processed. Crucially, the size semantics $[\![\mathrm{S}]\!]$ (eq. (2)) depend on the *running posterior* p_t as the effective context, making the threshold θ_k dynamically sensitive to what has already been established by previously interpreted adjectives.

1.4 Utterance prior (cost)

We incorporate two cost signals into the utterance prior, depending on the dataset:

(i) **Slider rating — subjectivity bias:** an additive penalty favouring size-first (S-initial) orderings, captured by a bias parameter b.

(ii) **Production — LM-derived cost:** $\mathrm{cost}_{\mathrm{LM}}(u) = -\beta \log P_{\mathrm{LM}}(u)$, where P_{LM} is a GPT-based prior over the 15 utterances and β controls sensitivity. This cost implicitly captures frequency and length biases: shorter, high-frequency forms (*e.g.* S) are cheaper.

1.5 Speaker models

Global speaker $S_{\mathbf{G}}$. A standard RSA pragmatic speaker that evaluates a complete utterance's utility:

$$P_{S_{\mathrm{G}}}(u \mid o, \mathcal{C}) \propto \exp\big[\alpha \cdot \log L_0(o \mid u, \mathcal{C}) - \lambda \cdot \mathrm{cost}(u)\big]. \tag{4}$$

Incremental speaker $S_{\mathbf{I}}$. Produces one adjective at a time; at step t it chooses the next adjective a_t given the prefix already produced. The updated-listener posterior after each step feeds into the next step's cost-weighted speaker distribution:

$$P_{S_{\mathrm{I}}}(a_1 \mid o, \mathcal{C}) \propto \exp\big[\alpha \cdot \log L_0(o \mid a_1, \mathcal{C}) - \lambda \cdot \mathrm{cost}(a_1)\big], \tag{5}$$

$$P_{S_{\mathrm{I}}}(a_t \mid a_{1:t-1}, o, \mathcal{C}) \propto \exp\big[\alpha \cdot \log p_t(o) - \lambda \cdot \mathrm{cost}(a_t \mid a_{1:t-1})\big]. \tag{6}$$

The joint probability of a complete utterance $u = (a_1, \ldots, a_T)$ factorises by the chain rule into a product of next-word decisions:

$$P_{S_{\mathrm{I}}}(u \mid o, \mathcal{C}) = \prod_{t=1}^{T} P_{S_{\mathrm{I}}}(a_t \mid a_{1:t-1},\, o,\, \mathcal{C}). \tag{7}$$

Intuition: how the incremental speaker works. The global speaker S_{G} scores all 15 complete utterances simultaneously, selecting the one that best balances informativeness and cost (eq. (4)). The incremental speaker S_{I}, by contrast, builds the utterance one adjective at a time from left to right. At step 1 it selects the first adjective that is most informative given the full context $\mathcal{C}$; the literal listener then updates its beliefs accordingly; at step 2 the speaker decides whether to produce an additional adjective, now evaluated against the listener's *updated* posterior rather than the original context. This sequential process models left-to-right production planning, where each adjective reshapes the communicative landscape for all subsequent ones (eqs. (6) and (7)).

Intuition: why ordering matters. The key mechanism is the context-sensitivity of the size semantics (eq. (2)): the threshold θ_k depends on the effective context at the point where S is evaluated. For the global speaker, this operates through right-to-left compositional semantics (eq. (3)): in SC, the listener first processes C (rightmost, closer to the noun), narrowing the posterior to colour-matching objects, before evaluating S over this filtered set. In CS, S is processed first over the full original context. For the incremental speaker, the same asymmetry arises through production order: when S is produced first, its informativeness is evaluated over the full context at step 1; when C is produced

first, S is evaluated at step 2 against the listener's colour-updated posterior. In both models, the *order* in which adjectives appear changes how informative each one is—and it is this asymmetry that drives ordering preferences as an emergent pragmatic phenomenon.

Intuition: why redundant adjectives occur. In both speaker models, each utterance's utility trades off informativeness against cost. A second adjective is produced whenever its marginal informativeness gain outweighs the additional cost. For the global speaker, this trade-off is evaluated jointly across all 15 utterances: a two-adjective form SC beats S alone when adding colour raises $\log L_0(o \mid u, \mathcal{C})$ enough to compensate for the longer utterance. For the incremental speaker, after the first adjective has been produced the listener's posterior p_1 may still assign non-negligible probability to distractors; if a second adjective reduces this residual uncertainty sufficiently, the speaker produces it even though the referent may already be probable. Both mechanisms predict *more* over-informative utterances when a single property is insufficient to uniquely pick out the referent (*e.g.*, the both-relevant condition) and *fewer* when one adjective is already highly informative (*e.g.*, sharp-size contexts)—consistent with the dominance of single-adjective S in sharp conditions and increased multi-adjective use in both-relevant conditions.

1.6 Model variants & parameterisation

Crossing the two speaker models (S_{G} vs. S_{I}) with two levels of participant-level pooling yields four model variants per dataset, compared via LOO-CV. The semantic parameters k, w_f, and ν are held fixed at theoretically motivated values (see below); inference targets the pragmatic and observation-level parameters.

Slider-rating models. The slider models compare only two utterances (big blue vs. blue big), using a subjectivity bias as the cost term and a Zero-One-Inflated Beta (ZOIB) observation model.

Model	Speaker	Participant pooling	Inferred parameters
Global, pooled	S_{G}	Single parameter set across all participants	α, b, σ
Global, hierarchical	S_{G}	Per-participant additive intercepts δ_i	α, b, σ, τ, δ_i
Incremental, pooled	S_{I}	Single parameter set across all participants	α, b, σ
Incremental, hierarchical	S_{I}	Per-participant additive intercepts δ_i	α, b, σ, τ, δ_i

Production models. The production models evaluate all 15 utterances with LM-derived costs. The hierarchical variants add per-participant deviations on α.

Model	Speaker	Participant pooling	Inferred parameters
Global, pooled	S_{G}	Single parameter set across all participants	α, β
Global, hierarchical	S_{G}	Per-participant deviations on α	α, β, τ, δ_i
Incremental, pooled	S_{I}	Single parameter set across all participants	α, β
Incremental, hierarchical	S_{I}	Per-participant deviations on α	α, β, τ, δ_i

Fixed semantic parameters (shared across all model variants): $k = 0.5$, $w_f = 1.0$, $\nu_{\text{colour}} = 0.8$ (slider and production), $\nu_{\text{form}} = 0.7$ (production only). Although not inferred, these parameters play a central role in shaping the model's predictions; figs. 1 to 3 illustrate their effects via the simulation (section 3).

Parameter glossary.

Parameter	Support	Interpretation
Inferred		
α	$\mathbb{R}_{>0}$	Pragmatic rationality: sharpness of the speaker's softmax over utterances
b	$\mathbb{R}_{>0}$	Subjectivity bias favouring S-initial orderings (slider models only)
σ	$\mathbb{R}_{>0}$	ZOIB dispersion controlling the spread of predicted slider ratings (slider models only)
β	$\mathbb{R}_{>0}$	Sensitivity to LM-derived utterance cost (production models only)
τ	$\mathbb{R}_{>0}$	SD of participant-level random effects (hierarchical variants only)
δ_i	$\mathbb{R}$	Per-participant intercept or deviation (hierarchical variants only)
Fixed (not inferred; effects shown in figs. 1 to 3)		
k	$(0, 1)$	Threshold position within the contextual size range; lower k = stricter threshold (fig. 1)
w_f	$\mathbb{R}_{\geq 0}$	Perceptual blur in the Weber-fraction size comparison; higher w_f = noisier size semantics (fig. 2)
ν	$(0, 1)$	Colour/form semantic value: probability that a feature-matching object is truly described by the adjective; higher ν narrows the size-first advantage (fig. 3)

Discussion: Challenges and experiences in fitting the models

- **Computational efficiency.** Making inference tractable required systematic vectorisation of data and parameters (leveraging JAX's batch semantics), JIT compilation of the entire model graph, and precomputing all quantities that do not depend on MCMC samples—in particular, marginalising out stochastic dependencies that could otherwise be recomputed at every leapfrog step.
- **Slow and stuck chains; chain diagnostics.** NUTS chains were frequently slow or got stuck in low-probability regions, making the usual workflow of running many short diagnostic chains prohibitively expensive. This motivated a dedicated posterior-analysis pipeline to detect problems early: visual inspection of posterior distributions for bad shapes, number of divergences, cross-chain disagreement, high autocorrelation within chains, and strong parameter correlations. Defining what counts as "successful inference" (*e.g.* by inspecting by-condition posterior predictives) helps a lot.
- **Slider rating data: bathtub-shaped distribution.** Slider ratings pile up at 0 and 1 (boundary values), violating standard continuous likelihood assumptions. We adopted a Zero-One-Inflated Beta (ZOIB) likelihood that explicitly models boundary probability mass via mixture components—a point mass at 0 with probability π_0, a point mass at 1 with probability π_1, and a Beta distribution for interior values with probability $1 - \pi_0 - \pi_1$.
- **Production data: degenerate and context-dependent distributions.** The free-production data exhibit highly concentrated, context-dependent categorical distributions—many utterance types receive near-zero counts in some conditions—making the full likelihood surface very steep and difficult to navigate. Our solution was to fix the semantic parameters (k, w_f, ν) at shared values across both datasets and infer only the pragmatic parameters (α, β) from the production data.

2 Experiments: Overall Design

The two experiments are complementary. The slider-rating study elicits graded acceptability judgements for pre-specified descriptions, framed from the speaker's perspective (*i.e.*, "how well does this description pick out the target?"), providing continuous data on how well each description fits a given context. The free-production study records the descriptions that speakers actually generate, yielding categorical choice data that reveal ordering preferences and the frequency of redundant modifiers. Together, the two tasks separate semantic fit (*how good is this description?*) from pragmatic choice (*which description does a speaker produce?*).

Both experiments share the same 3×2 factorial design:

Factor	Levels
F_1: **Relevant property**	Size-relevant, Colour-relevant, Both-relevant
F_2: **Sharpness**	Sharp (sizes well separated), Blurred (sizes clustered)
Fixed: Referent = largest, colour-matching, form-matching object in a 6-object context	

The adjective pair shown on each trial varied (size–colour, size–form, or colour–form), but because form and colour share the same semantic type—determinate features parameterised by ν—they are treated identically in the model. All analyses are collapsed across adjective-pair types. Each item encodes a context as a 6×3 matrix (s, c, f) with the referent at index 0 (fig. 4).

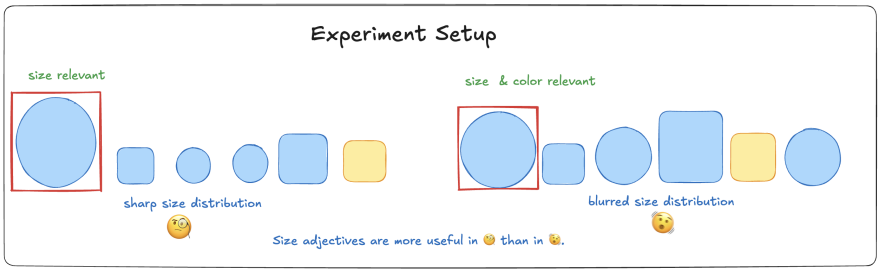

Figure 4: Illustration of an experiment item (placeholder). Left panel: a sharp-size context; right panel: a blurred-size context. The referent (circled) is the largest, colour-matching, form-matching object.

2.1 Slider-rating study (Experiment 1)

Task. Participants rated the *felicity* of each adjective description on a continuous slider (0–100) for a given context-referent pair. Ratings were collected for single-adjective (S, C, F) and two-adjective (SC, SF, CS, *etc.*) descriptions.

Dependent variable. Normalised slider ratings from [0, 100] to [-0.5, 0.5], mean-centred around 0, positive values indicating a preference for size-first orderings and negative values indicating a preference for colour/form-first orderings.

Discussion: Key findings (lmer: centred slider $\sim$ relevance $\times$ sharpness $+$ (1|participant) $+$ (1|item); figs. 5 and 6).

- **Relevance main effect** ($F(2, 3349) = 213.7$, $p < .001$; $\chi^2(2) = 402.6$, $p < .001$): size-first ordering preference is highest in size-relevant contexts, second highest in both-relevant contexts, and lowest in colour-relevant contexts.
- Even in colour-relevant contexts, mean centred ratings remain above zero; the reverse ordering (below zero) is qualitatively not preferred.
- **Sharpness main effect**: not significant ($F(1, 132) = 0.03$, $p = .86$).

- **Relevance × Sharpness interaction** ($F(2, 3349) = 5.84$, $p = .003$; $\chi^2(2) = 11.7$, $p = .003$): sharpness modulates the size-first advantage asymmetrically—it *increases* the preference in size-relevant contexts (blurred: 0.30, sharp: 0.34) but *decreases* it in both-relevant (blurred: 0.25, sharp: 0.20) and colour-relevant contexts (blurred: 0.06, sharp: 0.04).

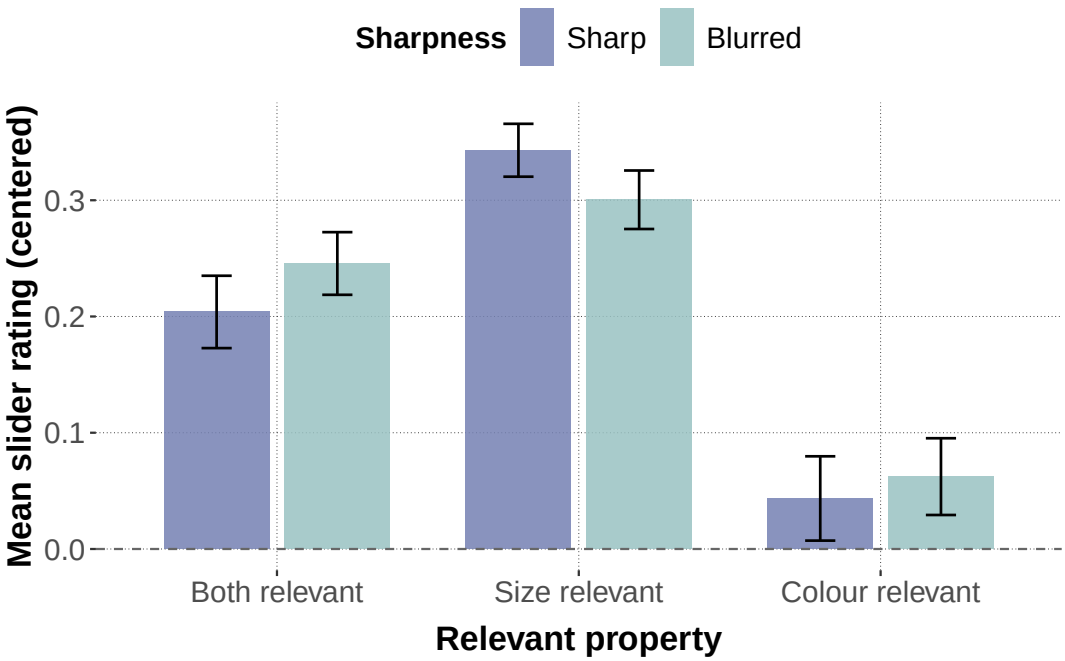

Figure 5: Slider-rating empirical results. Mean centred slider ratings (0 = no preference) by relevant-property condition and sharpness, with 95% CIs. Size-first orderings are preferred across all conditions, but the preference is strongest in size-relevant contexts and weakest in colour-relevant contexts. Sharpness modulates the size-first advantage asymmetrically across relevance conditions.

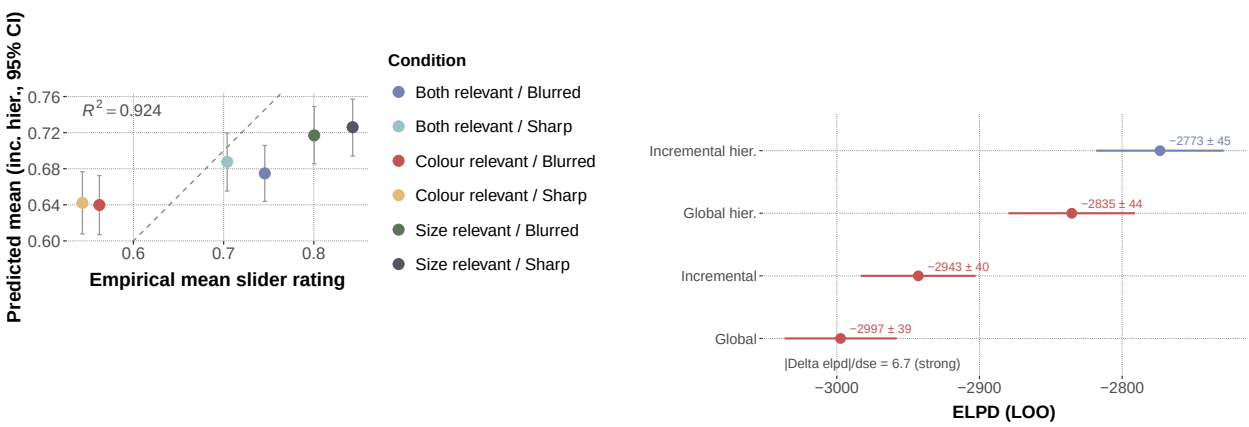

Figure 6: Left: Condition-level observed vs. predicted mean slider ratings for the incremental hierarchical model, with R^2 annotation and 95% CIs. Right: LOO-ELPD comparison across four model variants (best model highlighted in blue); $|\Delta\text{elpd}|/\text{dse} = 6.7$ (strong evidence for the incremental hierarchical model).

2.2 Free-production study (Experiment 2)

Task. Participants described the referent object to a hypothetical listener in as few words as possible, using only the provided adjectives. Responses were manually annotated into the 15 utterance categories.

Dependent variable. Categorical annotation $y_i \in \{1, \ldots, 15\}$ for each trial i.

Discussion: Key findings (glmer: binomial, (1|participant) + (1|item); figs. 7 and 8).

(1) Size-first preference (S-initial utterances coded as 1, all others as 0):

- **Relevance main effect** ($\chi^2(2) = 1859.7$, $p < .001$): size-first orderings dominate in size-relevant contexts; colour-first orderings dominate in colour-relevant contexts; both-relevant falls in between.
- **Sharpness main effect**: not significant ($\chi^2(1) = 0.79$, $p = .37$).
- **Relevance × Sharpness interaction** ($\chi^2(2) = 31.4$, $p < .001$): sharpness modulates the size-first advantage asymmetrically across relevance conditions—it reduces size-first preference in size-relevant contexts but increases it in both-relevant contexts.

(2) Over-informative utterances (three-adjective responses in both-relevant; multi-word responses in size-/colour-relevant coded as 1):

- **Relevance main effect** ($\chi^2(2) = 725.7$, $p < .001$): over-informative utterances are most frequent in size-relevant contexts, less frequent in both-relevant and colour-relevant contexts.
- **Sharpness main effect** ($\chi^2(1) = 10.3$, $p = .001$): over-informative utterances are more frequent in the blurred condition than in the sharp condition.
- **Relevance × Sharpness interaction** ($\chi^2(2) = 63.8$, $p < .001$): the sharpness effect on over-informativity is strongest in size-relevant contexts (blurred elicits far more multi-word utterances) and attenuated in both-relevant and colour-relevant contexts.

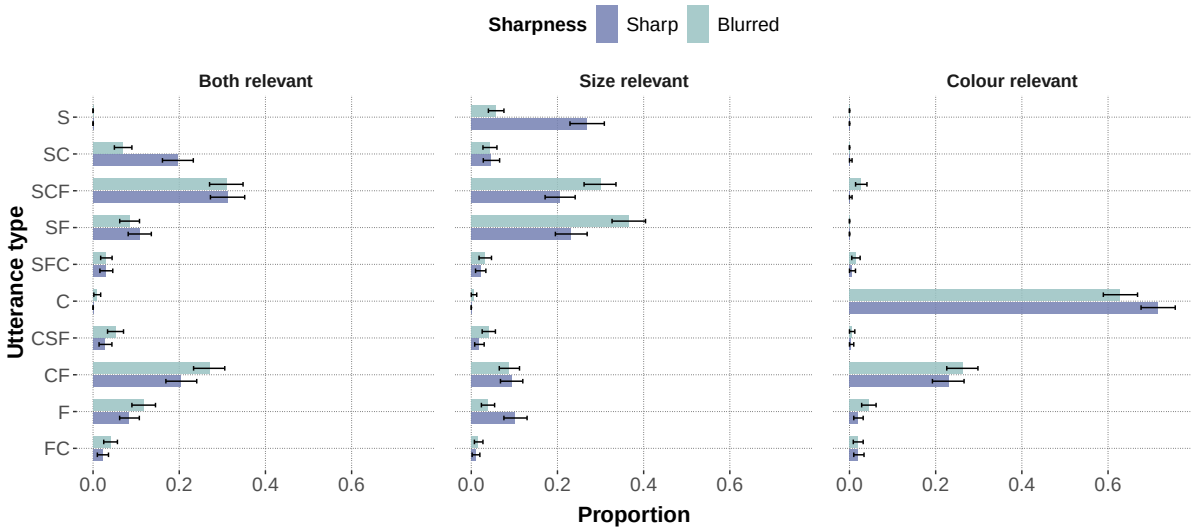

Figure 7: Production empirical results. Utterance-type proportions (with bootstrapped 95% CIs) by relevant-property condition and sharpness. Size-first orderings dominate in size-relevant contexts; colour-first orderings dominate in colour-relevant contexts. Only utterance types with $\geq 2\%$ in at least one condition are shown.

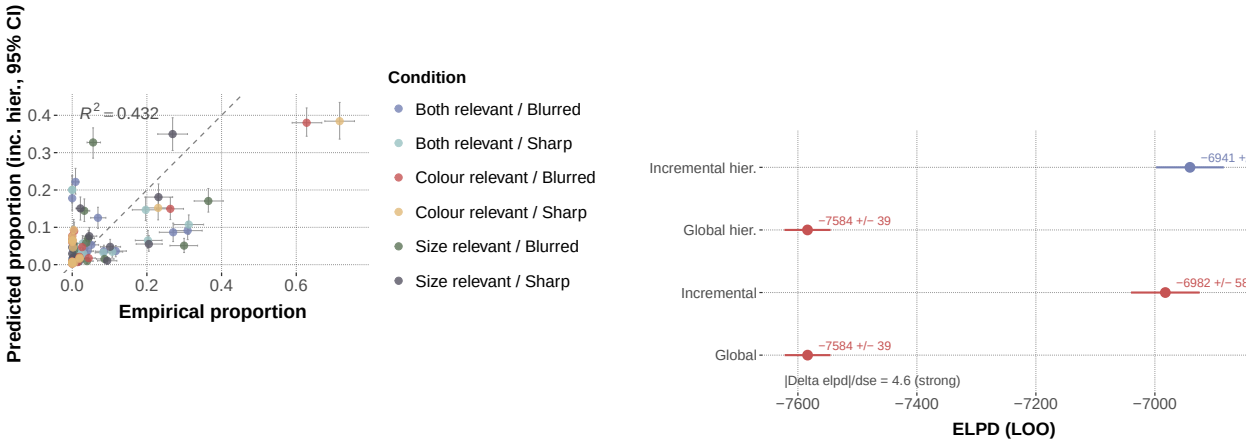

Figure 8: Left: Observed vs. predicted utterance-type proportions (per condition × utterance type) for the incremental hierarchical model, with R^2 annotation and 95% CIs. Right: LOO-ELPD comparison across four model variants; $|\Delta\text{elpd}|/\text{dse} = 4.6$ (strong evidence for the incremental hierarchical model).

3 Simulation

3.1 Goals

The simulation has three purposes:

(1) Establish that the incremental speaker *generates* ordering preferences, not just accommodates them, across a broad range of contexts.

(2) Map out which parameter region produces a qualitative *big-before-colour* preference and compare to a *colour-before-big* region.

(3) Assess robustness: does the ordering preference persist under varying context size (n_{obj}), size-distribution spread (σ_{spread}), and perceptual blur (w_f)?

3.2 Design

Parameter	**Values swept**	**# levels**
Speaker model	Incremental, Global	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν (colour semantic value)	0.90, 0.92, 0.94, 0.96, 0.98	5
k (threshold)	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f (perceptual blur)	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination		1000
Total rows		**6,000,000**

The parameter σ_{spread} controls the standard deviation of the truncated normal from which object sizes are drawn within a context. A small value (e.g. $\sigma_{\text{spread}} = 2$) produces *blurred* contexts in which objects are similar in size, making size adjectives less informative; a large value (e.g. $\sigma_{\text{spread}} = 15$) yields *sharp* contexts with well-separated sizes, making size adjectives highly informative.

Each simulated trial computes communicative success $p(o_{\text{ref}} \mid u)$ for two key orderings (big blue > blue big) and records which ordering the speaker model assigns higher probability.

3.3 Results

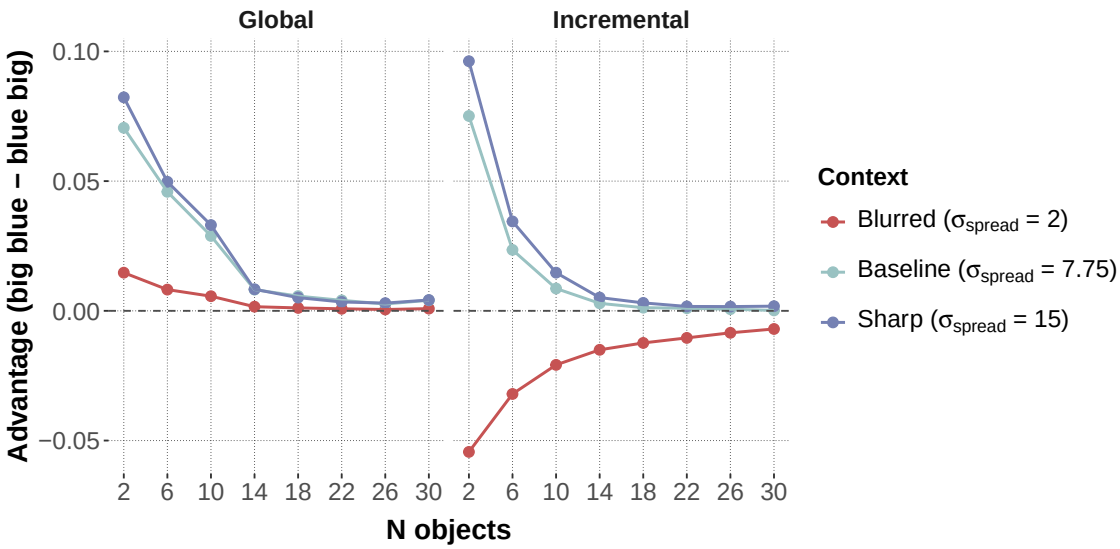

Figure 9: Size-first advantage (`big blue` − `blue big`) as a function of context size n_{obj} and context sharpness σ_{spread}, faceted by speaker type. Both speakers show a positive advantage in sharp contexts that diminishes with increasing context size. Crucially, the incremental speaker exhibits a *negative* advantage (colour-first preference) in blurred contexts with few objects, a pattern absent from the global speaker.

Discussion: Simulation findings

- Both speaker types generate a size-first ordering advantage across the majority of the parameter space, demonstrating that the emergence of ordering preferences is not exclusive to the incremental mechanism.
- The global speaker derives its ordering asymmetry solely from context-dependent, sequential-update compositional semantics—i.e., the way the literal listener composes meaning by updating beliefs object-by-object. This is the *minimal mechanism*: it requires no production-level incrementality, only a compositional semantics that is sensitive to contextual informativity. This is compatible with previous listener-oriented communicative-efficiency accounts of subjectivity-based ordering preferences. While those accounts relied on non-agent simulations, our explicit RSA model captures, extends, and confirms their core insight within a fully agent-based framework.
- The incremental speaker amplifies this effect and additionally produces qualitatively distinct predictions (e.g., a *reversal* to colour-first preference in blurred, small-context conditions; fig. 9), suggesting that the incremental production mechanism contributes an additional, separable source of ordering preferences beyond the minimal compositional-semantics baseline.

4 General Discussion

4.1 Summary

We presented an RSA model with two speaker variants—global and incremental—that derives adjective ordering preferences from the interaction of context-sensitive compositional semantics, communicative efficiency, and, in the incremental case, left-to-right production planning. The model is fitted to two complementary behavioural datasets (slider ratings, Experiment 1; free production, Experiment 2) via NUTS-based Bayesian inference in NumPyro / JAX, and its qualitative predictions are stress-tested across a 6-million-trial parameter simulation.

Discussion: Key findings

- **Ordering preferences emerge from compositional semantics alone.** Even the global speaker—which has no production-level incrementality—generates a size-before-colour preference across most of the parameter space, because applying a color adjective first to the hierarchical compositional semantics narrows the comparison class and thereby boosts the informativeness of a subsequent size adjective (section 3).
- **Incremental production amplifies and qualitatively extends the effect.**
- Also **incremental hierarchical model fits the data best.**
- **Q: Do we need an incremental speaker model?**

Discussion: Outlook

- What is the right framing for the paper: pragmatics, psycholinguistics, or cognitive science?
- Target venue? **[TODO: CogSci / Cognition / PLOS ONE / Glossa / ?]**
- What level of detail should each section have for the target audience?

References

[TODO: Add references]